

AI Tutor

Product Requirements Document

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1. Introduction

We are building an AI-powered tutor to teach primary school children mathematics. The tutor uses the parent's own AI provider (API key) to generate personalised learning content, knowledge checks, and diagnostic feedback — adapting continuously to each child's demonstrated ability.

The initial release covers Mathematics only. Additional subjects will be scoped after the mathematics implementation is validated.

2. General Principles

- The tutor interleaves teaching with frequent knowledge checks, in a variety of formats:
 - Multiple-choice questions
 - Free-text answer boxes, with answers analysed by LLM
 - Teaching should be Socratic: students work things out themselves rather than being told answers directly. For example, rather than stating that demand slopes downward, the tutor asks what happens to demand as price falls.
 - Knowledge checks and quizzes are tailored to the student's current ability level. A student who consistently answers correctly should receive harder problems.
 - When a student answers incorrectly, the tutor asks follow-up questions to elicit their reasoning, then uses an LLM to parse that reasoning and explain clearly where they went wrong. This applies in both Learn and Practice modes.
 - Students may ask questions at any point during a lesson via a dedicated question button, which opens a chat-style interface.
 - The system continuously tracks answers and progress to adapt content and difficulty over time.
 - Diagrams should be generated as structured, renderable text output (e.g. SVG) where they aid understanding. No raster images. No animations.
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3. Design

3.1 Basic UI

- Web-based application, accessible on laptops and tablets.
- Clean, modern design. Background colour: #F5F5F5 throughout.
- Home page offers two actions: Log In and Create Account.

3.2 Creating Accounts & Logging In

3.2.1 Parent Accounts

- Parents sign up with an email address and password.
- During sign-up, parents specify:
 - Their preferred AI model
 - Their API key for that model
- The API key is stored in local storage only — it is never transmitted to or stored on any server.
- If a parent logs in from a new device, they will be prompted to re-enter their API key.

3.2.2 Child Accounts

- Parents create child accounts from the Parent Dashboard.
- Required fields when creating a child account:
 - Username
 - First name
 - Year of birth

3.2.3 Login Routing

- On login, children are directed to the Children's Dashboard.
- On login, parents are directed to the Parent Dashboard.

3.3 Parent Dashboard

- Option to create a new child account, always visible.
- For each child, the following is displayed:
 - Child's name and total points
 - AI-generated performance summary (see Section 5.3)
 - Expandable rows by subject:
 - Collapsed view: subject name and status label

N/A — child has not engaged with the subject

Below expectations

On track

Above expectations

- Expanded view: AI-generated subject summary (strengths and areas of difficulty), followed by module-level rows showing which modules have been studied and the child's performance in each
- Settings area including AI model selection and API key entry.

3.4 Children's Dashboard

- Displays the child's total points prominently.
 - Navigation to available subjects.
 - For a new subject, only a 'Let's get started!' button is shown until the initial assessment is complete.
 - Once the assessment is complete, available modules are shown. Each module displays Learn and Practice options.
 - Points: children earn one point per correct answer in any knowledge check or practice session. Points are purely motivational. A rewards system is a planned future extension.
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4. Backend

4.1 Data Storage

- Firebase is used for all user data storage.
- Data stored per child includes: account details, all session history, all question answers (correct and incorrect), module completion status, current ability level, and running practice averages.

4.2 AI Provider

- The system uses the parent's own AI provider and API key for all AI-generated content.
 - The API key is held in local storage on the parent's device and passed with each API request. It is never stored server-side.
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5. Child Assessment

5.1 Ability Scale

Each child's ability is tracked on an 8-point scale, updated after every completed Learn or Practice session. The scale is used to tailor content difficulty and to generate parent-facing summaries.

Level	Qualitative Description	Parent Dashboard Label
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1	The child is significantly behind the expected level for their age.	Below expectations
2	The child is behind the expected level for their age, but not concerningly so.	Below expectations
3	The child is in line with the bottom 50% of their age group.	On track
4	The child is in line with the top 50%–75% of their age group.	On track
5	The child performs in the top 25% of their age group.	On track
6	The child performs in line with the average of children about one year older than them.	Above expectations
7	The child performs in line with the top 25% of children about one year older than them.	Above expectations
8	The child performs in line with children about two years older than them.	Above expectations

5.2 Level Calculation Formula

The child's level is recalculated after every completed Learn or Practice session. The formula is incremental — only two values need to be stored (EMA_score and answer count n), not the full answer history.

Each question generated by the AI is tagged with an expected_accuracy value (0–1): the probability that a typical child of that age answers it correctly. Performance is measured relative to this expectation, so a child who correctly answers a hard question gains more than one who answers an easy one.

```
// Per answer update (incremental EMA, no history required):
adjusted_answer = answer - expected_accuracy
  where answer = 1 (correct) or 0 (incorrect)
  and expected_accuracy is AI-supplied per question (0-1)

EMA_new = w * EMA_old + adjusted_answer

Store only: EMA_score, n (total answers)

// Level mapping (EMA_score -> Level):
below -0.30      -> Level 1
-0.30 to -0.18  -> Level 2
-0.18 to -0.06  -> Level 3
-0.06 to 0.06   -> Level 4
0.06 to 0.18    -> Level 5
0.18 to 0.30    -> Level 6
0.30 to 0.42    -> Level 7
above 0.42      -> Level 8
```

The EMA_score is centred around 0, where 0 means performing exactly as expected for the child's age. Positive scores indicate above-expectation performance; negative scores indicate below. Only EMA_score and n are persisted — no individual answer history is required.

5.3 AI-Generated Parent Summary

- The parent-facing summary is generated fresh every time the Parent Dashboard is opened.
- The AI reads the child's full stored history — all answers, session records, module completions, and current level — and produces an accurate, non-sycophantic summary.
- The summary must reflect actual performance. It should not default to positive framing if the data does not support it.

6. Curriculum: Mathematics

6.1 Module Sequence

Modules are gated linearly: a module unlocks only when the previous module is complete. The 'Let's get started' assessment can unlock a student partway into the sequence, bypassing modules they have already effectively demonstrated mastery of.

Module completion criterion: an EMA score of +0.06 or above (Level 4 or higher, i.e. performing at or above age expectation) with a minimum of 10 practice questions answered in that module.

#	Module	Approximate Age Range (UK)
1	Counting	Ages 5–6 (Year 1)
2	Addition	Ages 5–6 (Year 1)
3	Subtraction	Ages 5–6 (Year 1)
4	Simple multiplication	Ages 6–7 (Year 2)
5	Simple division	Ages 6–7 (Year 2)
6	Fractions	Ages 7–8 (Year 3)
7	Basic shapes	Ages 7–8 (Year 3)
8	Area	Ages 8–9 (Year 4)
9	Volume	Ages 9–10 (Year 5)
10	Long division	Ages 9–10 (Year 5)
11	Place value & decimals	Ages 9–10 (Year 5)
12	Negative numbers	Ages 9–10 (Year 5)
13	Percentages	Ages 10–11 (Year 6)

14	Long multiplication	Ages 10–11 (Year 6)
15	Ratio & proportion	Ages 10–11 (Year 6)
16	Basic algebra	Ages 11–12 (Year 7)
17	Linear equations	Ages 11–12 (Year 7)
18	Angles	Ages 11–12 (Year 7)
19	Perimeter	Ages 11–12 (Year 7)
20	Advanced area	Ages 11–12 (Year 7)
21	Statistics & data	Ages 12–13 (Year 8)
22	Probability	Ages 12–13 (Year 8)
23	Coordinate geometry	Ages 12–13 (Year 8)
24	Sequences & patterns	Ages 12–13 (Year 8)
25	Powers, roots & prime factorisation	Ages 12–13 (Year 8–9)

6.2 Let's Get Started (Initial Assessment)

- When a child opens Mathematics for the first time, the only available action is 'Let's get started!'.
- Tapping this triggers a series of AI-generated multiple-choice questions designed to gauge the child's existing knowledge across the curriculum.
- Based on performance, the system places the child at an appropriate point in the module sequence, unlocking all modules up to and including their assessed starting point.
- The assessment also sets the initial difficulty calibration for each unlocked module.
- Modules beyond the child's assessed level remain locked and unlock sequentially as the child completes each one.

Let's Get Started Prompt

You are an AI tutor assessing a child's current mathematics ability.
The child was born in {year_of_birth}.

Generate a sequence of multiple-choice questions to assess their knowledge across the following mathematics topics, in order:
{ordered_module_list}

Rules:

- Start with questions appropriate for the youngest age group and increase difficulty.
- Ask enough questions per topic to make a reliable placement judgment.
- Stop a topic's questions early if the child is clearly not yet at that level.
- Each question must have exactly 4 answer options, with one correct answer.
- Questions should be clear, unambiguous, and age-appropriate in language.
- Do not explain answers during the assessment.

Output format: JSON array of question objects:

```
[
  {
    "module": "<module name>",
    "question": "<question text>",
    "options": ["A", "B", "C", "D"],
    "correct": "<correct option letter>",
    "expected_accuracy": <float 0-1, probability a typical child of this age
answers correctly>
  }
]
```

6.3 Learn Mode

- Each module has a lesson plan generated once when the module first unlocks.
- The lesson plan is generated using the prompt below. It is not regenerated unless the module is reset.
- Lessons are Socratic: they guide students to work things out themselves.
- Teaching should rely on structured questions and reasoning, not direct explanation.
- Text-based diagrams (e.g. SVG) should be included where they aid understanding.
- No raster images. No animations.
- Students may ask questions at any point via the question button (chat interface). These questions are answered by the AI in context, then the lesson resumes.
- Knowledge checks are embedded throughout. When a student answers incorrectly, a follow-up diagnostic conversation is triggered (see Section 6.5).

Lesson Plan Generation Prompt

You are an AI tutor teaching mathematics to a child.

Child profile:

- Year of birth: {year_of_birth}
- Current ability level: {level} out of 8 (see scale below)
- Modules already completed: {completed_modules}
- Demonstrated strengths: {strengths}
- Areas of difficulty: {weaknesses}

Ability scale:

- 1 = significantly behind expected level for age
- 2 = behind, but not concerningly so
- 3 = bottom 50% of age group
- 4 = top 50-75% of age group
- 5 = top 25% of age group
- 6 = average of children ~1 year older
- 7 = top 25% of children ~1 year older
- 8 = level of children ~2 years older

Task: Generate a lesson plan to teach: {module_name}

Principles to follow:

- Be Socratic: guide the student to work things out themselves. Do not state facts directly; ask questions that lead the student to the answer.
- Teach concepts and vocabulary first, then interrogate with knowledge checks.

- Calibrate difficulty to the child's level. A level 1 child needs simpler language and easier problems than a level 6 child.
- Embed knowledge checks throughout (not only at the end).
- Where a diagram would help understanding, generate it as SVG.
- Use age-appropriate language throughout.
- Do not use garlic in any word problems involving food.

Output: A structured lesson plan as a JSON object with sections, each containing teaching steps and embedded knowledge check questions.

6.4 Practice Mode

- Practice sessions generate a fresh set of questions each time the student starts a new session.
- Questions are calibrated to the student's current ability level and their performance history in that module.
- The practice prompt is used at the start of every new practice session.
- Module completion requires: a practice EMA score of +0.06 or above, with at least 10 questions answered in that module.
- EMA_score and answer count (n) are stored in Firebase and updated after each session. No individual answer history is stored.

Practice Session Generation Prompt

You are an AI tutor generating a practice session for a child.

Child profile:

- Year of birth: {year_of_birth}
- Current ability level: {level} out of 8
- Module being practised: {module_name}
- Cumulative practice EMA score in this module: {ema_score} (scale: -1 to +1, 0 = performing as expected for age)
- Demonstrated strengths in this module: {module_strengths}
- Areas of difficulty in this module: {module_weaknesses}

Task: Generate a set of 10 practice questions for {module_name}.

Rules:

- Mix question formats: multiple-choice and short free-text answers.
- Calibrate difficulty to the child's level. If EMA score is above +0.15, increase difficulty. If below -0.15, reduce difficulty.
- Focus more questions on demonstrated areas of weakness.
- Questions should be varied and not repetitive across sessions.
- Use age-appropriate language.

Output format: JSON array of question objects:

```
[
  {
    "type": "multiple_choice" | "free_text",
    "question": "<question text>",
    "options": ["A", "B", "C", "D"], // omit for free_text
    "correct": "<correct answer or option letter>",
    "expected_accuracy": <float 0-1, probability a typical child of this age
answers correctly>
```



```
}  
]
```

6.5 Diagnostic Follow-up (Wrong Answers)

- When a student answers a question incorrectly in either Learn or Practice mode, a diagnostic conversation is triggered.
- The tutor asks the student to explain their reasoning in their own words (free-text input).
- The student's explanation is passed to the LLM, which identifies the specific misconception and explains clearly what went wrong and why the correct answer is right.
- The explanation is framed constructively and at the child's reading level.
- After the explanation, the student may be given a similar question to attempt again.

7. Future Considerations

The following are explicitly out of scope for the initial release but should be designed around from the start:

- Additional subjects beyond Mathematics.
- Rewards system linked to the points total.
- Multi-device support: if the API key is ever moved server-side, full multi-device login becomes possible without re-entry.
- COPPA / GDPR-K compliance: mandatory before any release beyond personal use.
- Cost optimisation: parent dashboard summaries are regenerated on every open; at scale, caching triggered by meaningful events (module completions, level changes) would reduce API costs significantly.
- Raster image support if a suitable curated or generated image pipeline can be established.